

The Future of Active Directory Security

 fidelissecurity.com/threatgeek/active-directory-security/the-future-of-active-directory-security

Neeraja Hariharasubramanian

May 27, 2024



Understanding Active Directory

Active Directory is the central directory service created by Microsoft which helps manage users, computers, and other resources within a Microsoft Windows network. It is used to track devices, access and passwords of each employee and store in one central vault.

The biggest advantage of Active Directory is it allows organizations to customize data organization, manage user accounts, and control access through group policies. With AD, users enjoy single sign-on, while administrators benefit from easy resource management and robust security features.

AD's scalability, [data protection](#), and automatic updates make it an essential component for Windows-based networks. Remember to plan carefully when implementing AD in your organization.

But these very advantages are what makes it an extremely lucrative target for cyber attackers.

Safeguarding Active Directory in the Era of Cyber Threats

Dive into the depths of Active Directory threats and security:

- Current landscape of Active Directory
- Active Directory threats to beware of
- Proactive Strategies to Protect AD

[Watch Now](#)

Evolution of Active Directory

Since its inception in 1999, the [Microsoft Active Directory](#) landscape has evolved rapidly. Designed to provide centralized access and a hierarchical framework for managing resources within a network, AD made its way to becoming the cornerstone of IT infrastructures.

However, with the arrival of cloud architectures and businesses increasingly moving towards cloud environments, AD had to evolve. That is how Azure AD, a cloud-based identity and access management service, came into being.

Difference: Active Directory vs Azure AD

Traditional AD was created to address the challenges of organizations operating within a localized network environment. It includes features like:

Centralized Management: With AD, administrators can control access and permissions to network resources like PCs, printers, and user accounts all from one place.

- **Group Policy:** With the help of this capability, operating systems, apps, and user settings in an AD environment can be managed and configured centralized.
- **Security Abilities:** Kerberos-based authentication is integrated with AD, offering a strong foundation for protecting user credentials and resource access.

While this traditional AD model worked well for on-premise AD environments, a newer version was needed to keep up with the technological advancements and newer requirements for identity management solutions.

[Azure Active Directory](#) came into existence in 2010, which is a more evolved version of traditional AD, meeting the demands of cloud-based and hybrid environments. These capabilities include:

Integrated to Cloud: Azure AD offers multi-factor authentication (MFA) and single sign-on (SSO) for hundreds of SaaS apps, including [Microsoft 365](#), and is built to integrate easily with cloud services.

- **Ability to Scale:** Unlike on-prem AD, Azure AD can scale drastically to accommodate the needs of global enterprises without the need for massive and comprehensive physical infrastructure.
- **Identity Protection:** Cloud resources are more securely protected thanks to the integrated [identity threat detection and response](#) capabilities of Azure AD that aid in preventing identity theft.

Here is the basic comparison of Active Directory vs Azure AD:

METRIC		
Deployment	On-Premises, Own Infrastructure	Cloud-based, No local servers needed
Focus	Manages on-premise resources	Manages resources cloud-first
Security	Own Internal Vigilance	Constant Updates by Microsoft's Security
Scalability	Requires Additional Hardware	No Additional resource requirements

Future of Active Directory & It's Security

The cloud-first approach of many enterprises would make one think that the future of Active Directory is Azure AD only. But that is not the case. In the current landscape of things, most enterprises still can't survive without a local AD infrastructure.

This indicates that while Active Directory continues to serve as a critical component of on-premises identity management, Azure AD represents the future, offering scalable, secure, and flexible identity and access management solutions for cloud and hybrid environments.

This means that the future of likely lies in a hybrid approach. It is expected to become a mixed bag, with a blend of on-premises AD coexisting with cloud-based Azure AD for some time, eventually transitioning more towards the cloud.

The hybrid security approach enables:

Unique Identity Management: Consolidated and streamlined identity management for usage in cloud and on-premises settings, resulting in a more seamless administrative experience and user experience.

Enhanced Security Posture: Combining Azure AD's sophisticated, cloud-specific security features—like conditional access and identity protection—with the strong security features of classic AD can help enhance the security posture.

Future developments in the field of Active Directory security will include advanced security features to keep up with the evolving cyber threat landscape.

Zero Trust Architecture: The implementation of a [Zero Trust security architecture](#), in which every step of the process requires verification and trust is never assumed. To guarantee safe resource access, this model will be incorporated into AD environments.

Artificial Intelligence & Machine Learning: The combination of AI and [machine learning results in improved threat detection](#) and response capacities. Large volumes of data can be analyzed by these technologies in real time to spot unusual activity and possible security risks which is imperative for the future.

Multi-Factor Authentication: Increased MFA adoption and use is likely to add an additional layer of protection to the authentication process and lessen the dependence on passwords alone.

Apart from this, here are some other developments that are likely to shape the world of Active Directory Security.

Increased use of automation and orchestration: Automation will likely be leveraged for quicker [threat detection and response](#). It will also help with policy enforcement with automated scripts detecting policy infringement.

Integration with newer technologies: Potential integration with technologies like blockchain and biometric is possible soon. These will make processes more secure and user-friendly along with supporting accurate identity verification processes.

Cloud first & mobile first approach: Managing mobile devices is going to take over the future of Active Directory Security. This combined with [securing for cloud-native applications](#) and services will be pivotal in ensuring seamless access and management.

Monitoring and Compliance: Compliance with regulatory standards like [GDPR](#), HIPAA, and [CCPA](#) is going to continue to remain the focus. Emphasis on real-time detection will continue for which continuous monitoring will be required.

Fidelis Active Directory Intercept

With Active Directory Security, as important as it is to detect threats quickly, it is also vital to response to these threats with the same speed. And that is where [Fidelis Security](#) comes

in.

[Fidelis Active Directory Intercept™](#) is the only solution which combines AD-aware [network detection and response](#) (NDR) with [integrated deception technology](#) with foundational AD log and event monitoring. It has the ability to:

Catch [Active Directory risks](#) coming from multiple sources

Get complete visibility and insights into your organization's resources

Enables security teams to seamlessly catch configuration issues caused due to ongoing changes in Microsoft Active Directory

Gain complete context and information about the attacks

Low privileged permissions are required as information is fetched using Lightweight Directory Access Protocol (LDAP)

If you want a complete AD security solution to protect your enterprise's crown jewels, then [get in touch with our experts today!](#)

This datasheet dives into how Active Directory Intercept works. It covers:

- Network Traffic Analysis
- Active Directory Log and Event Monitoring
- Intercept and Defeat AD Attacks

[Download Datasheet](#)

[Talk to Expert](#)